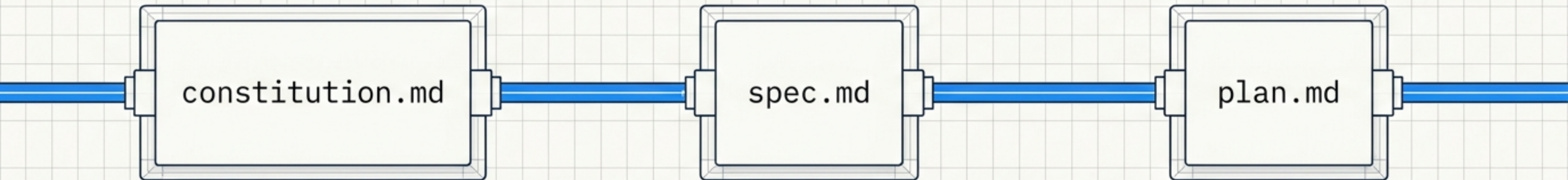
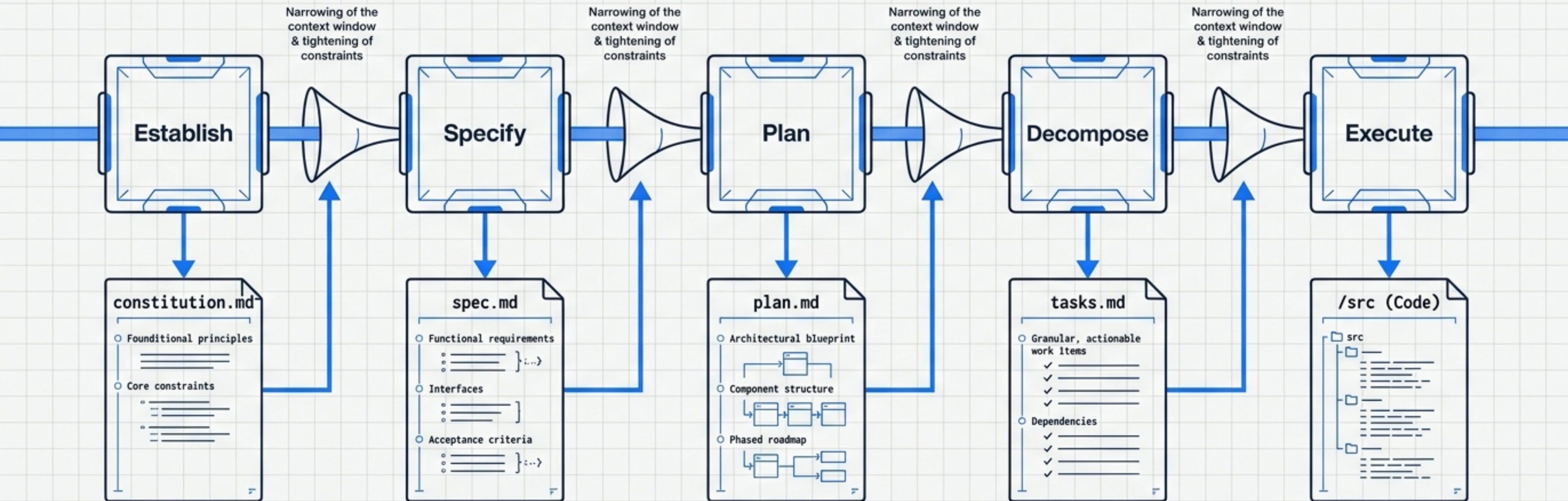


Engineering Intent: The SDD Operational Playbook

A blueprint-to-execution reference for Spec-Driven Development workflows and artifacts.



The Assembly Line of Intent: Systematically Eliminating Ambiguity



Phase 1: Embedding Immutable Laws in constitution.md

constitution.md

- # Foundational principles

- # Architectural Baselines

- # Directory Conventions

Security by Construction

Explicit CWE (Common Weakness Enumeration) vulnerability mappings and non-negotiable security requirements.

Architectural Baselines

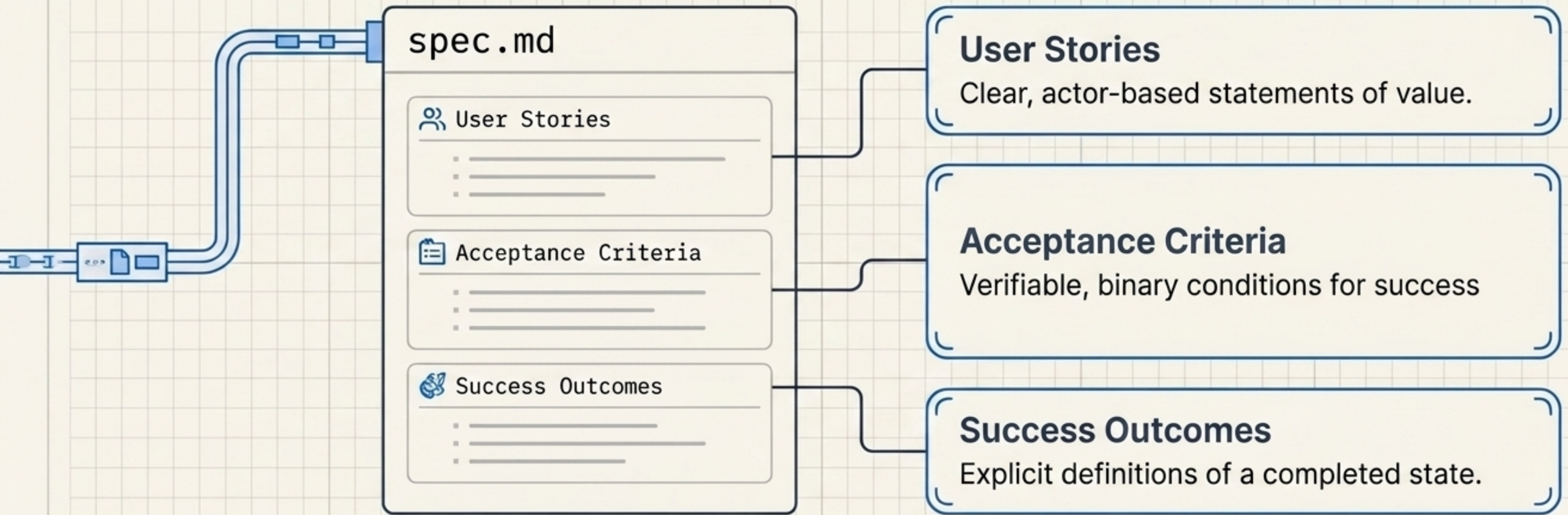
Immutable tech-stack laws.

Directory Conventions

Hardcoded rules for file structures and naming conventions.

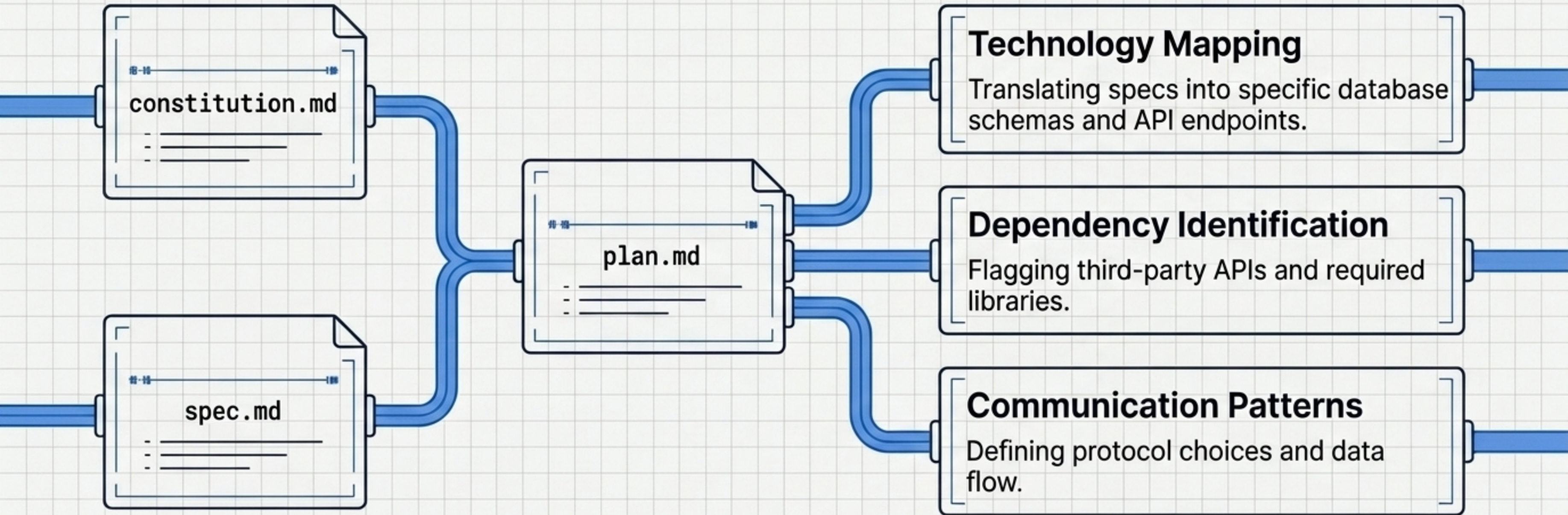
The Constitution sits at the apex of the hierarchy. It prevents the erosion of human authority by ensuring AI agents never suggest banned libraries or vulnerable patterns.

Phase 2: Defining Functional Intent in spec.md



Focus exclusively on the What and Why. Avoid pseudo-code or technical implementation details here to prevent Markdown Madness and over-constraining the AI's problem-solving capabilities.

Phase 3: Architecting the Execution via plan.md



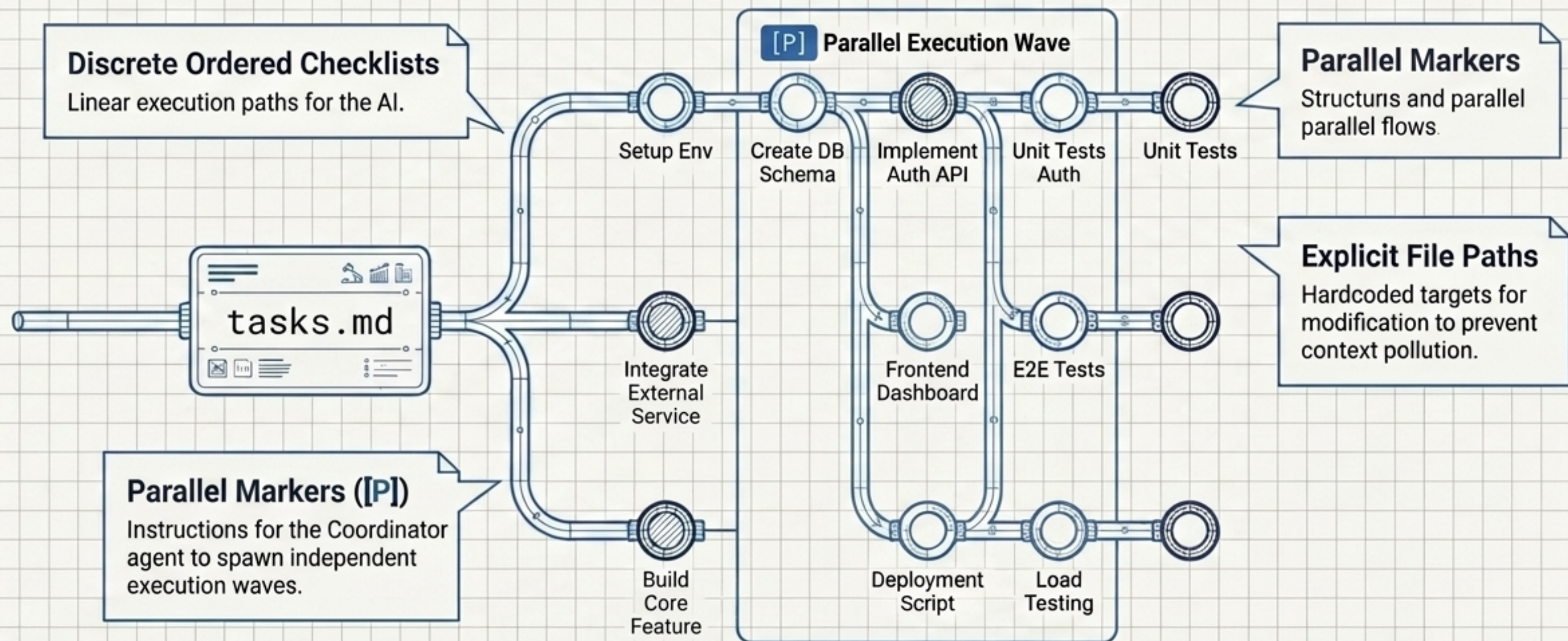
The plan is where functional intent meets architectural reality. It acts as the primary technical context for all downstream generation.

Decision-Driven Software Engineering (DDSE): The TDR Matrix

TDR Type	Focus Area	Impact on AI Workflow
MDD (Major Design Decision)	Strategic choices (e.g., Monolith vs. Microservices)	Global constraints applied to every agent session.
ADR (Architectural Decision Record)	System architecture and design	Defines structural patterns (e.g., Event-driven, Repository pattern).
EDR (Engineering Decision Record)	Development practices and tooling	Standardizes tooling and linting practices.
IDR (Implementation Decision Record)	Component-level code choices	Rationale for specific, function-level logic choices.
CDR (Contract Decision Record)	API and integration contracts	Enforces cross-service compatibility and API schemas.
TDM (Tech Decision Memo)	Evaluations of rejected alternatives	Lightweight matrix to prevent AI rework loops.

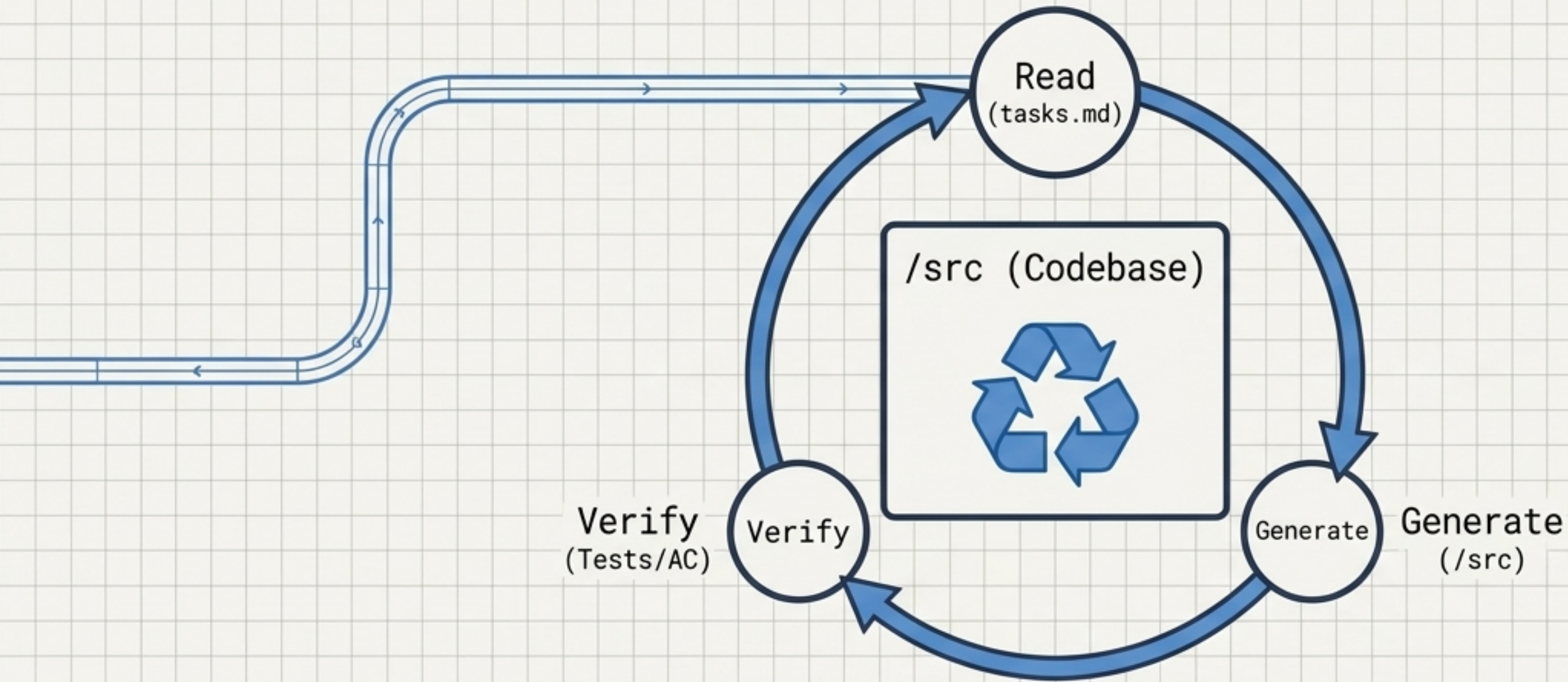
Technical Decision Records (TDRs) use YAML frontmatter to create decision memory, ensuring stateless AI agents respect historical architectural choices.

Phase 4: Atomic Task Decomposition in tasks.md



Monolithic complexity breaks AI context windows. `tasks.md` isolates work into testable, reviewable chunks optimized for multi-agent orchestration.

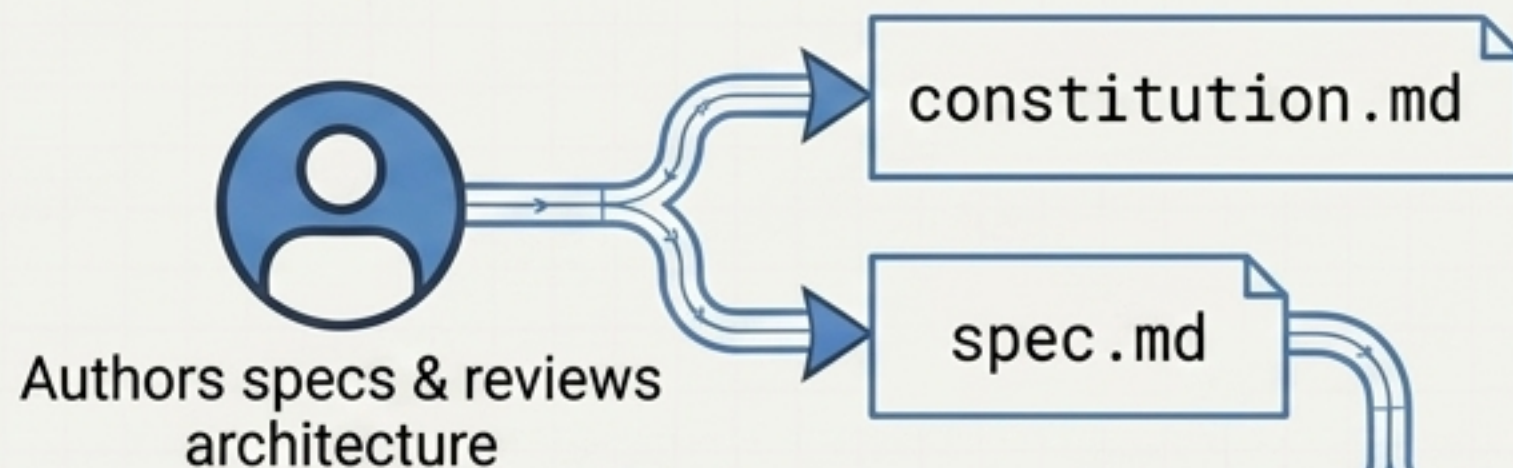
Phase 5: The Read-Write-Verify Implementation Loop



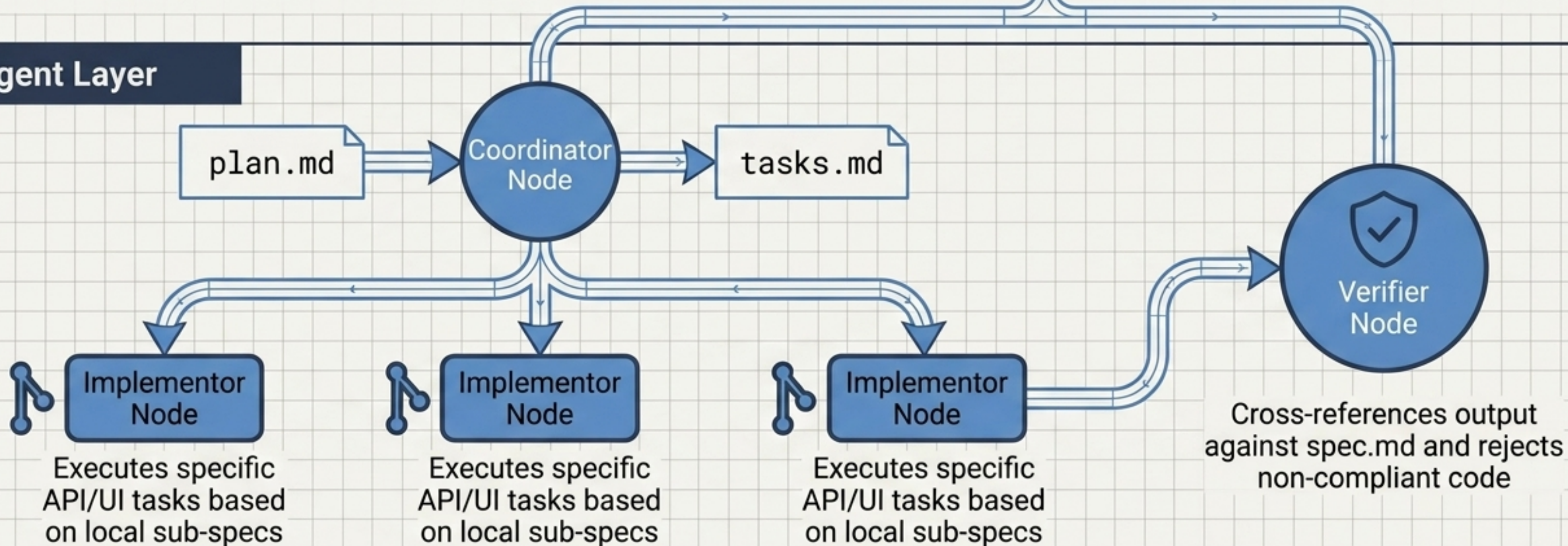
Code is a generated, disposable artifact. The Source of Truth Rule: If a constraint requires a change in direction, update the specification, not the code. Divergences must be resolved upstream to maintain perfect synchronicity.

The Adversarial Orchestration Model: Multi-Agent Execution

Human Layer

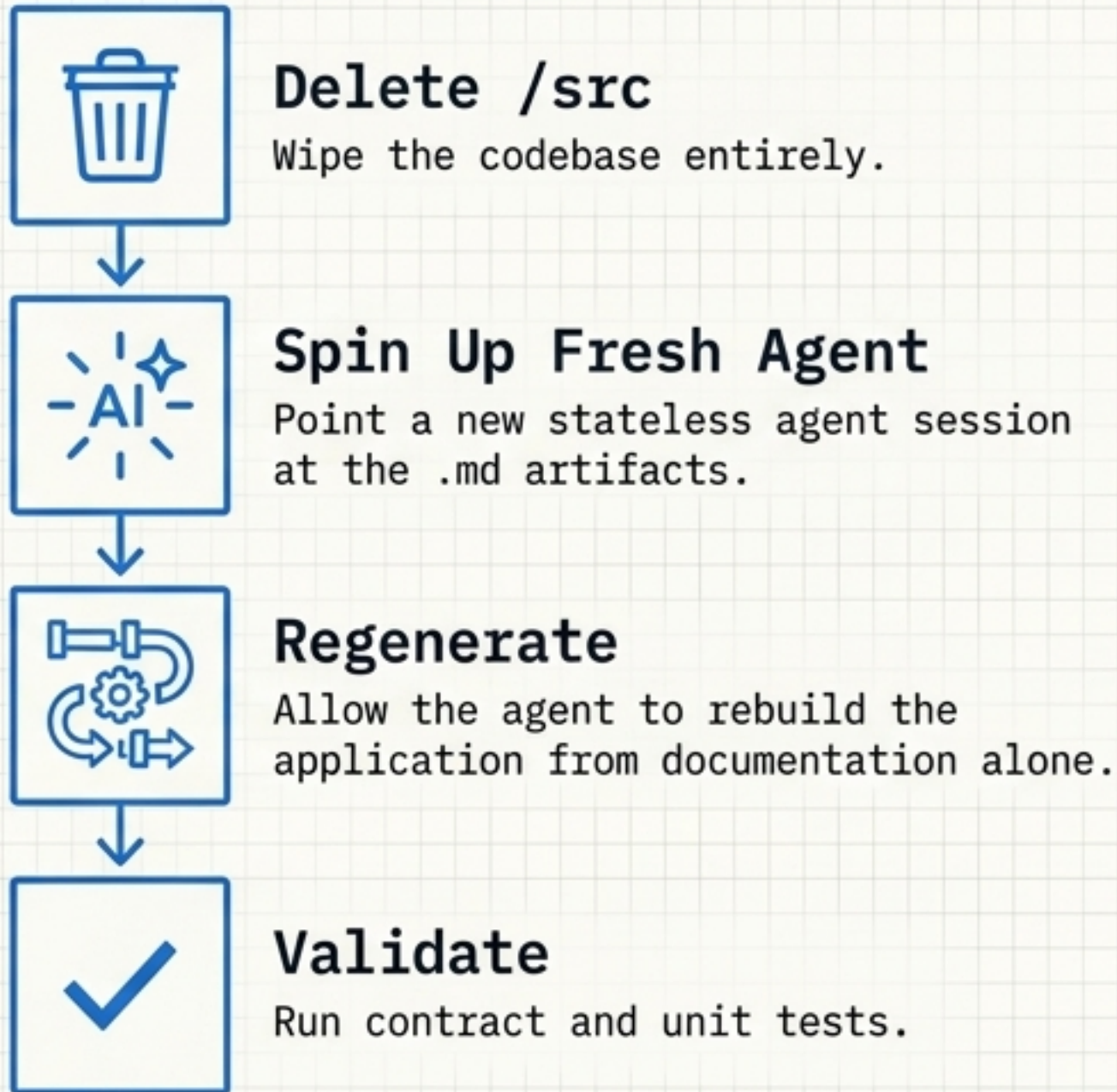


AI Agent Layer

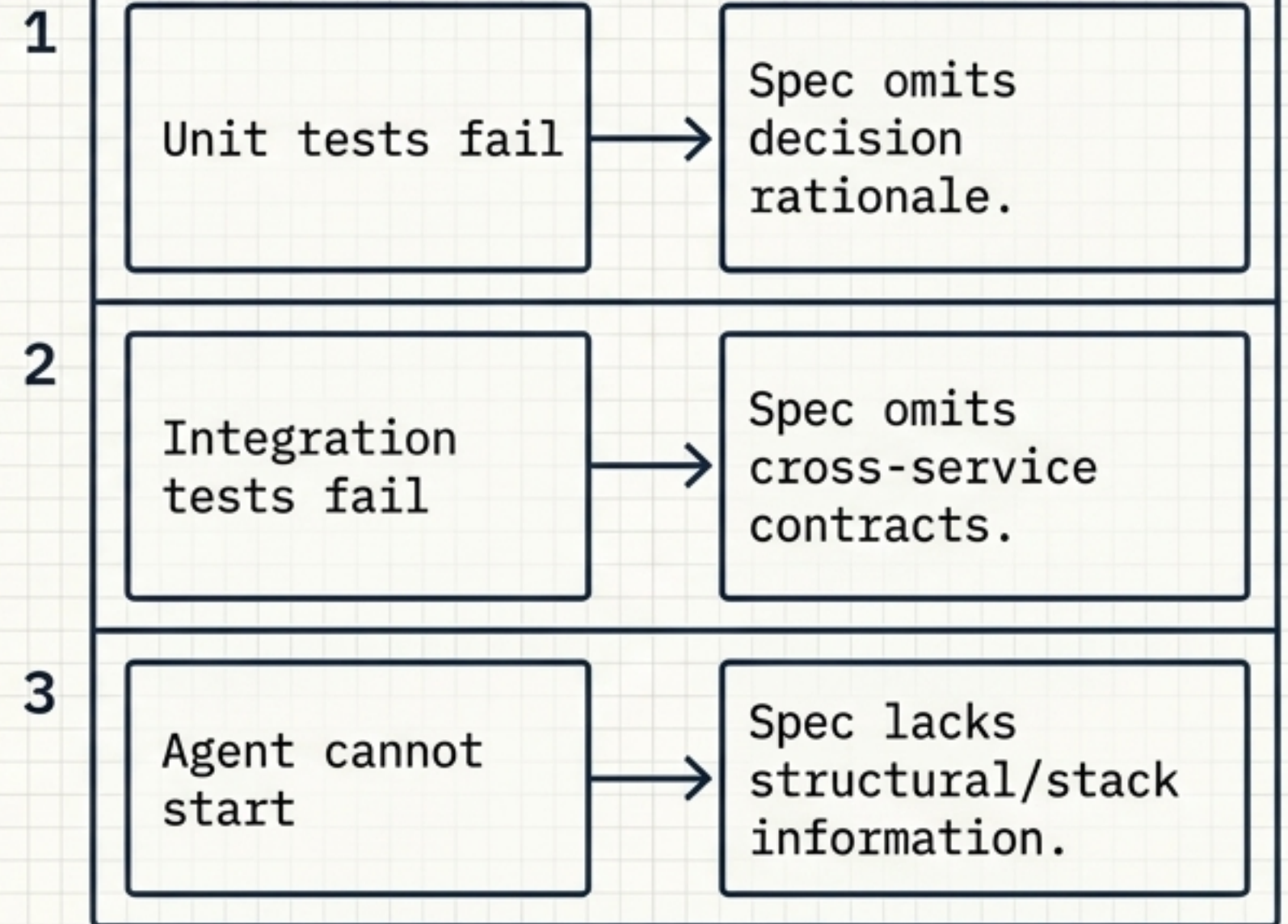


Artifact Traceability and The Rebuild Test

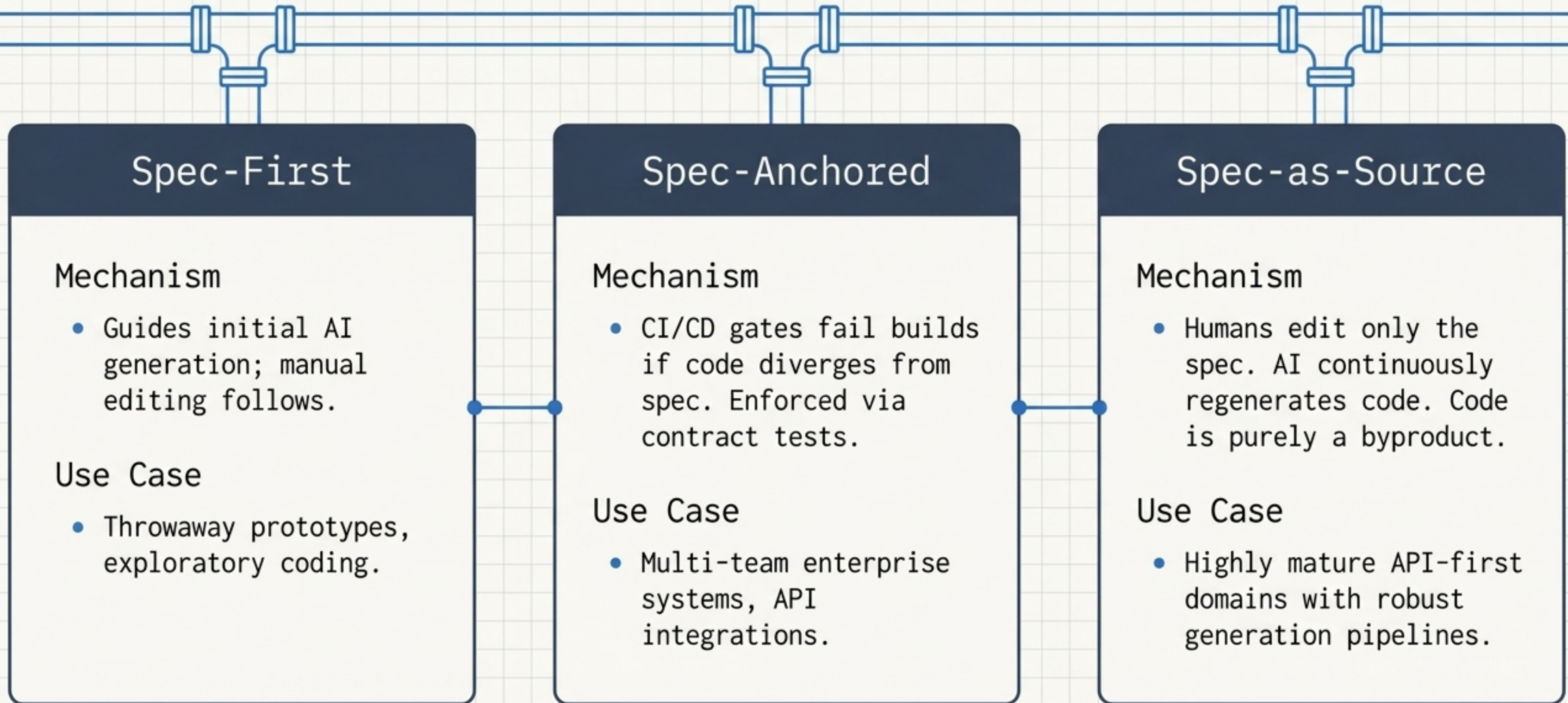
THE REBUILD TEST FLOW



FAILURE DIAGNOSTICS



Application Spectrums: Matching Rigor to Requirements



General Conclusions

Shift in Human-AI Roles

- Humans focus on defining intent and architectural constraints.
- AI acts as the execution engine for generation and testing.
- Requires new skills in prompt engineering and spec-writing.

Balancing Rigor and Speed

- Different application types demand varying levels of rigor (Spectrum).
- Choose the right approach: Spec-First, Spec-Anchored, or Spec-as-Source.
- Traceability is key to maintaining quality across all levels.

Artifacts as Source of Truth

- Documentation (specs, plans, constitution) becomes primary.
- Code is a generated byproduct, validated against the artifacts.
- Enable workflows like the 'Rebuild Test' to ensure consistency.